

Part 1 of 2

Android Application: Series RLC Circuit

K. BASKAR

An RLC circuit consists of a resistor, a capacitor and an inductor connected in a series or parallel to each other. Such circuits can be used in oscillator circuits, filter circuits, radio receivers and others. This article describes a series RLC and a parallel circuit with an Android application. Using this application, you can plot the graph for capacitive reactance, inductive reactance, impedance, admittance, bandwidth, phase angle, inductive susceptance, capacitive susceptance, current and voltage. The application uses Achartengine version 1.1.0 graph library to simulate graphical output.

Series RLC circuit

In the series RLC circuit, the resistor (R), inductor (L) and capacitor (C) are connected in a series with an AC input supply. In the resistor, voltage and current are in phase with input AC supply; in the inductor, voltage leads the current by 90° ; and in the capacitor, voltage lags the current by

90° . An example of the series RLC circuit used in an Android app is shown in Fig. 1. The example output of the series RLC circuit is shown in Fig. 2.

Series RLC circuit at resonance.

Resonance occurs if the inductor and capacitor are present in the circuit. Here, energy is transferred back and forth between elements and, at a



particular frequency, the two energies are equal. This frequency is called resonance frequency. The ideal-case inductive reactance versus frequency is shown in Fig. 4. The plotted graph shown in Fig. 5 is based on the inputs as shown in Fig. 1.

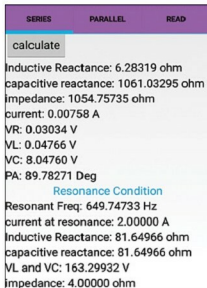


Fig. 2: Example output of the series RLC circuit in an Android app

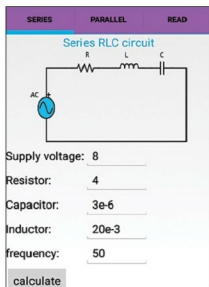


Fig. 1: Example of the series RLC circuit in an Android app



Fig. 3: Series circuit output at resonance

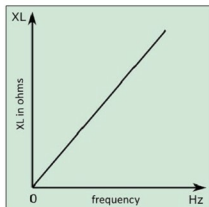


Fig. 4: Ideal case of inductive reactance versus frequency plot

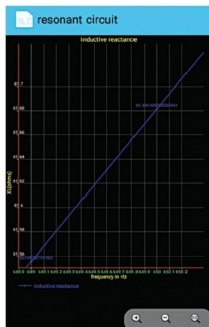


Fig. 5: Inductive reactance versus frequency plot in Android